

CASCADE-CLASSICAL CORRESPONDENCE IV: DISCRETE CLOSURE DYNAMICS

ABSTRACT. This paper puts dynamics on the refinement-spaces substrate of paper P3. At each refinement level $n \geq 0$ and branch $\bullet \in \{H_4, D_4\}$, we define an explicit quadratic closure functional

$$F_n = \alpha R_n + \beta E_n - \gamma Q_n = \frac{1}{2} \langle \cdot, L_n \cdot \rangle$$

on the finite-level Hilbert space $X_n := X_n^{0,\bullet} \oplus X_n^{1,\bullet}$, where R_n, E_n, Q_n are the curvature, kinetic, and mass quadratic forms built from the discrete coboundary, graph Laplacian, and mass operator of §2. The gradient of F_n is represented by a bounded self-adjoint operator $L_n \in B(X_n)$ (Theorem 3.3); the continuous-time gradient flow $\dot{\Phi}_n = -L_n \Phi_n$ has unique global solutions $\Phi_n(t) = e^{-tL_n} \Phi_n(0)$ (Theorem 4.1); energy dissipates along the flow as $d_t F_n(\Phi_n(t)) = -\|L_n \Phi_n(t)\|^2 \leq 0$ (Proposition 5.1).

A discrete-time Euler scheme $\Phi_{n,k+1} = (I - \eta L_n) \Phi_{n,k}$ is analysed in Section 6. Energy monotonicity under small step size is separated cleanly from norm contractivity (Theorem 6.6), which holds only on coercive invariant subspaces Y_n of X_n where L_n has a positive spectral gap.

Refinement compatibility of the full gradient operator $L_n = \alpha A_n + \beta B_n - \gamma C_n$ is established here only in the mass-only case $\alpha = \beta = 0$ (Theorems 7.6 and 7.7); the two constituent results that are proved are the factor- $\frac{1}{2}$ coboundary relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ (Proposition 7.2) and the exact mass-block compatibility $p_{n+1,n} C_{n+1} = C_n p_{n+1,n}$ (Proposition 7.4). For arbitrary (α, β, γ) this paper makes no claim about refinement- intertwining of L_n (neither positive nor negative); see Remark 7.5. Downstream papers needing refinement-intertwining of cascade dynamics in that regime must supply their own analysis.

The false global theorem “all correspondences are Borel / equivariant” from an earlier foundations draft is replaced by an explicit explicit generator-family list of *admissible finite-level intertwiners* (Definition 8.2), with printed operator-norm bounds and continuity/Borel measurability following directly from finite-dimensional functional calculus (Theorem 8.4). This is the fourth of thirteen planned cascade infrastructure papers.

No continuum-limit, Gromov-Hausdorff, or manifold-identification claim is made here. No target-side metric, hydrodynamic, gauge, spectral, Hecke, cohomology, or CTM correspondence theorem is in scope; those are P5-P13.

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1. INTRODUCTION AND SOURCE AUDIT

1.1. The repair target. The cascade framework has long referred to “closure dynamics” in heuristic language: a closure functional $\mathcal{F} = \alpha R + \beta E - \gamma Q$ whose critical points or gradient flows organise the cascade rungs. The earlier draft of the cascade correspondence foundations manuscript invoked this at an abstract global level via a “Theorem” asserting that every correspondence arising from closure dynamics is Borel and symmetry-equivariant (iter-1 review of that manuscript, Theorem 4.3 of its §4). The review identified that claim as false in the form stated, and recommended replacement by case-by-case operator-level bounds.

This paper executes that replacement at the finite-level stage. We work exclusively on the finite-dimensional Hilbert spaces $X_n := X_n^{0,\bullet} \oplus X_n^{1,\bullet}$ of paper P3, define the closure functional as an explicit quadratic form, identify its gradient as a bounded self-adjoint operator L_n , and derive its dynamics rigorously. Every operator appearing in downstream constructions (bonding maps, flow maps, symmetry actions, sector projections, coboundaries, adjoints, Laplacians, mass operators, and the specific compositions used by P5–P13) is given an explicit norm bound.

1.2. What this paper proves.

- (1) The closure functional $F_n = \alpha R_n + \beta E_n - \gamma Q_n$ is a continuous symmetric quadratic form on X_n , equivalently $F_n(\Phi) = \frac{1}{2} \langle \Phi, L_n \Phi \rangle$ for a unique bounded self-adjoint operator L_n (§3, Theorem 3.3).
- (2) The continuous-time gradient flow $\dot{\Phi}_n = -L_n \Phi_n$ has unique global solutions on $[0, \infty)$ given by $\Phi_n(t) = e^{-tL_n} \Phi_n(0)$, a finite-dimensional C^0 -semigroup on X_n (§4, Theorem 4.1).
- (3) Energy dissipation along the flow: $d_t F_n(\Phi_n(t)) = -\|L_n \Phi_n(t)\|^2 \leq 0$ (§5, Proposition 5.1).
- (4) Discrete-time Euler iteration: energy-decreasing under a step-size bound; strict norm-contraction only on coercive invariant subspaces of L_n (§6, Propositions 6.2 and Theorem 6.6).
- (5) Refinement compatibility for the mass-only case ($\alpha = \beta = 0$): the identity $p_{n+1,n} L_{n+1} = L_n p_{n+1,n}$ and the flow intertwining $p_{n+1,n} e^{-tL_{n+1}} = e^{-tL_n} p_{n+1,n}$ hold (§7, Theorems 7.6 and 7.7). The constituent intermediate identities proved are the factor- $\frac{1}{2}$ coboundary relation (Proposition 7.2) and the exact mass-block compatibility (Proposition 7.4). For arbitrary (α, β, γ) this paper makes no claim about refinement-intertwining of L_n ; see Remark 7.5.
- (6) Admissible finite-level intertwiners (a finite list of generators: bonding maps, cylindrical embeddings, sector projections, symmetry actions, coboundaries, adjoints, Laplacians,

L_n , e^{-tL_n} , and fibre maps from P2) have explicit operator-norm bounds; all admissible intertwiners are bounded, continuous, and Borel (§8, Theorem 8.4).

1.3. What this paper does not prove. To forestall any reader from treating P4 as a continuum-dynamics or branch-specific-correspondence paper, we record the exclusions explicitly:

- (1) No continuum-limit statement: no Gromov–Hausdorff or spectral-convergence claim about the $n \rightarrow \infty$ behaviour of F_n, L_n, e^{-tL_n} , or the bonding maps. Those belong to paper P6.
- (2) No continuum PDE identification. This paper gives no heat-equation, Ricci-flow, Navier–Stokes, or Yang–Mills-type interpretation of $\dot{\Phi}_n = -L_n\Phi_n$. Those are P5, P7, and P9.
- (3) No target-side metric, hydrodynamic, gauge, spectral, Hecke, cohomology, or CTM/TM correspondence theorem.
- (4) No universal “all correspondences are Borel and equivariant” global theorem. Instead Section 8 gives a explicit generator-family list with explicit bounds for the operators later papers actually invoke.
- (5) No claim that F_n has a unique minimiser or that the flow converges to the ground state without further spectral hypotheses. Energy monotonicity and norm contractivity are separated theorems with separate hypotheses.
- (6) No claim about uniqueness or classification of the closure functional at the level of abstract $W^\bullet$ -symmetric quadratic forms. (The cascade-foundations note F2 contains such a claim; P4 does not rely on or reproduce it.)

1.4. Source audit. Theorem-grade inputs to this paper:

Internal cascade infrastructure.: Paper P2 (cascade algebraic substrate [1]) for the sector fibres $V_{H_4}, V_{D_4}, V_{16}, V_8, \text{Im}(\mathbb{O})$ and their fixed trace forms. Paper P3 (cascade refinement spaces [2]) for the finite-level Hilbert spaces $X_n^{0,\bullet}, X_n^{1,\bullet}$, the downward bonding maps $p_{n+1,n}$, the adjoint cylindrical embeddings $j_{n,n+1}$, and the finite-level Coxeter actions (P3 only constructs the vertex inverse-limit Hilbert space and defines σ on the mixed $\mathbb{Q}(\varphi)/\mathbb{Q}$ vertex form, not on the real Hilbert space; we use σ here only at the rational-form level).

Classical mathematics.: Spectral graph theory and finite-graph Hodge theory [6]; bounded self-adjoint operators and spectral calculus in finite dimension [4, 5]; linear ODEs with bounded coefficients [7, 8].

We do not use as theorem-grade input any cascade-foundations F2 uniqueness claim, any iter-1 foundations.tex Theorem 4.3 global theorem, or any continuum-limit cascade note.

1.5. Organization. Section 2 fixes the finite-level discrete operators (coboundary, Laplacian, mass) and proves their basic bounds. Section 3 defines the closure functional and identifies its gradient. Section 4 constructs the continuous-time flow. Section 5 derives the energy dissipation identity. Section 6 analyses the discrete-time Euler scheme and separates energy monotonicity from norm contractivity. Section 7 establishes refinement intertwining for the coboundary block (factor $\frac{1}{2}$) and a mass-only refinement intertwining for L_n in the specialization $\alpha = \beta = 0$, and records the obstruction to claiming more in the general $\alpha, \beta, \gamma > 0$ regime. Section 8 replaces the false global Borel/equivariance theorem by explicit finite-generator bounds. Section 9 states the citation contract for downstream papers. Appendix A is a notation concordance.

2. FINITE-LEVEL STATE SPACES AND DISCRETE OPERATORS

2.1. State spaces.

Definition 2.1 (Combined state space). For $n \geq 0$ and $\bullet \in \{H_4, D_4\}$, let

$$X_n := X_n^{0,\bullet} \oplus X_n^{1,\bullet}$$

denote the direct sum of the vertex-sector and edge-sector ℓ^2 -spaces from [2, Defs. 4.2, 4.3], with the direct-sum inner product $\langle (f, A), (g, B) \rangle_{X_n} := \langle f, g \rangle_{X_n^0} + \langle A, B \rangle_{X_n^1}$. The *domain* is

$D_n := X_n$; since X_n is finite-dimensional (Proposition 2.2), the domain is automatically dense in itself and every operator introduced below is bounded.

Proposition 2.2 (X_n is finite-dimensional). *X_n is a finite-dimensional real Hilbert space with $\dim_{\mathbb{R}} X_n = \dim_{\mathbb{R}} X_n^{0,\bullet} + \dim_{\mathbb{R}} X_n^{1,\bullet} < \infty$. The dimension bounds are given by [2, Proposition prop:Xn-Hilbert]: $\dim_{\mathbb{R}} X_n^{0,\bullet} = |V(G_n^\bullet)| \cdot 32$ and $\dim_{\mathbb{R}} X_n^{1,\bullet} = |E(G_n^\bullet)| \cdot 7$, both finite.*

Proof. Direct from [2, Proposition prop:Xn-Hilbert]. $\square$

2.2. Discrete coboundary and adjoint.

Definition 2.3 (Coboundary operator). The *discrete coboundary* $d_n: X_n^{0,\bullet} \rightarrow X_n^{1,\bullet}$ is defined by

$$(d_n f)(e) := f(\text{head}(e)) - f(\text{tail}(e))$$

for $e \in E^{\text{or}}(G_n^\bullet)$ a level- n oriented edge, where $\text{head}(e), \text{tail}(e) \in V(G_n^\bullet)$ are its endpoints (with orientation). Here $f(\text{head}(e)) - f(\text{tail}(e))$ is computed in the fibre F^0 of [2, Notation 3.3], and the difference is then projected to $F^1 = \text{Im}(\mathbb{O})$ via the canonical quotient $F^0 \rightarrow F^0/(\text{non-Im}(\mathbb{O}) \text{ components}) \rightarrow \text{Im}(\mathbb{O})$ (i.e., taking the $V_8 = \mathbb{O}$ component, then projecting to its imaginary part).

d_n satisfies the antisymmetry convention on oriented edges: $(d_n f)(-e) = f(\text{head}(-e)) - f(\text{tail}(-e)) = f(\text{tail}(e)) - f(\text{head}(e)) = -(d_n f)(e)$, so $d_n f \in X_n^{1,\bullet}$ (which incorporates the antisymmetry convention of [2, Definition def:Xn-1]).

Definition 2.4 (Adjoint coboundary). The *adjoint coboundary* $d_n^*: X_n^{1,\bullet} \rightarrow X_n^{0,\bullet}$ is the ℓ^2 -adjoint of d_n with respect to the inner products of $X_n^{0,\bullet}$ and $X_n^{1,\bullet}$: for $A \in X_n^{1,\bullet}$ and $v \in V(G_n^\bullet)$,

$$(d_n^* A)(v) = \frac{1}{2} \sum_{e: v=\text{head}(e)} A(e) - \frac{1}{2} \sum_{e: v=\text{tail}(e)} A(e),$$

where each sum runs over oriented edges with v as head or tail respectively. The factor $\frac{1}{2}$ compensates for the $\frac{1}{2}$ in the X_n^1 inner product convention [2, Definition def:Xn-1].

Lemma 2.5 (Coboundary-adjoint is the inner-product adjoint). $\langle d_n f, A \rangle_{X_n^1} = \langle f, d_n^* A \rangle_{X_n^0}$ for all $f \in X_n^{0,\bullet}, A \in X_n^{1,\bullet}$.

Proof. Direct computation. Writing out both sides using Definitions 2.3 and 2.4 and the inner product conventions: the matching of sums over endpoints is standard discrete-Hodge book-keeping [6, §1.2] adapted to the antisymmetry convention. $\square$

2.3. Block operators on X_n .

Definition 2.6 (Graph Laplacian block A_n). The *graph Laplacian block* $A_n: X_n \rightarrow X_n$ is

$$A_n := \begin{pmatrix} d_n^* d_n & 0 \\ 0 & d_n d_n^* \end{pmatrix},$$

acting on $X_n = X_n^{0,\bullet} \oplus X_n^{1,\bullet}$. The top-left block $d_n^* d_n$ is the classical combinatorial graph Laplacian on X_n^0 (positive semi-definite by construction); the bottom-right block $d_n d_n^*$ is the edge-Laplacian on X_n^1 (also positive semi-definite).

Definition 2.7 (Coboundary block B_n). The *coboundary block* $B_n: X_n \rightarrow X_n$ is the self-adjoint extension of the off-diagonal coboundary-adjoint pair:

$$B_n := \begin{pmatrix} 0 & d_n^* \\ d_n & 0 \end{pmatrix}.$$

B_n is self-adjoint by Lemma 2.5. Its square is $B_n^2 = \begin{pmatrix} d_n^* d_n & 0 \\ 0 & d_n d_n^* \end{pmatrix} = A_n$, so B_n is a “square root”-type structure for the Hodge Laplacian.

Definition 2.8 (Mass block C_n). The *mass block* $C_n: X_n \rightarrow X_n$ is

$$C_n := \begin{pmatrix} \mathbf{P}_{V_{D_4}} & 0 \\ 0 & \text{id}_{X_n^1} \end{pmatrix},$$

where $\mathbf{P}_{V_{D_4}}$ is the orthogonal projection on $X_n^{0,\bullet}$ onto the V_{D_4} -sector component of the fibre F^0 (so $\mathbf{P}_{V_{D_4}}$ acts pointwise on each vertex value by the linear projection $F^0 = V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8 \rightarrow V_{D_4}$). C_n is self-adjoint and positive semi-definite.

Proposition 2.9 (Bounds on the block operators). *For every $n \geq 0$:*

- (1) d_n, d_n^* are bounded linear maps with $\|d_n\|_{\text{op}}, \|d_n^*\|_{\text{op}} \leq 2\sqrt{\deg(G_n^\bullet)}$, where $\deg(G_n^\bullet)$ is the maximum vertex degree of $G_n^\bullet$.
- (2) A_n is bounded self-adjoint with $0 \leq A_n \leq 4\deg(G_n^\bullet) \cdot \text{id}$.
- (3) B_n is bounded self-adjoint with $\|B_n\|_{\text{op}} \leq 2\sqrt{\deg(G_n^\bullet)}$.
- (4) C_n is bounded self-adjoint with $0 \leq C_n \leq \text{id}$.

All four operators are bounded in operator norm uniformly on X_n (but not uniformly in n , since $\deg(G_n^\bullet)$ may grow with refinement).

Proof. (1) *Coboundary bound.* For $f \in X_n^{0,\bullet}$,

$$\|d_n f\|_{X_n^1}^2 = \frac{1}{2} \sum_e \|f(\text{head}(e)) - f(\text{tail}(e))\|_{\text{Im}(\mathbb{O})}^2.$$

By the Cauchy–Schwarz inequality applied to the projection onto $\text{Im}(\mathbb{O})$ and the triangle inequality, $\|f(\text{head}(e)) - f(\text{tail}(e))\|^2 \leq 2(\|f(\text{head}(e))\|^2 + \|f(\text{tail}(e))\|^2)$. Summing over e :

$$\|d_n f\|^2 \leq \sum_e (\|f(\text{head}(e))\|^2 + \|f(\text{tail}(e))\|^2) \leq 2\deg(G_n^\bullet) \cdot \|f\|^2,$$

using that each vertex v appears as head or tail of at most $\deg(G_n^\bullet)$ edges. Hence $\|d_n\|_{\text{op}} \leq \sqrt{2\deg(G_n^\bullet)} \leq 2\sqrt{\deg(G_n^\bullet)}$. The same bound for $d_n^* = (d_n)^*$ follows from $\|T^*\|_{\text{op}} = \|T\|_{\text{op}}$ for bounded operators on Hilbert spaces.

(2), (3) *Laplacian and coboundary block bounds.* A_n is positive semi-definite as a sum of T^*T -type blocks (both $d_n^* d_n$ and $d_n d_n^*$ are T^*T or TT^*). The upper bound $A_n \leq 4\deg(G_n^\bullet) \text{id}$ follows from $\|d_n^* d_n\|_{\text{op}} \leq \|d_n\|_{\text{op}}^2 \leq 4\deg(G_n^\bullet)$.

For $B_n = \begin{pmatrix} 0 & d_n^* \\ d_n & 0 \end{pmatrix}$:

$$\|B_n(f, A)\|^2 = \|d_n^* A\|^2 + \|d_n f\|^2 \leq \|d_n\|_{\text{op}}^2 (\|A\|^2 + \|f\|^2) = \|d_n\|_{\text{op}}^2 \|(f, A)\|^2,$$

so $\|B_n\|_{\text{op}} \leq \|d_n\|_{\text{op}} \leq 2\sqrt{\deg(G_n^\bullet)}$.

(4) *Mass block bound.* $\mathbf{P}_{V_{D_4}}$ is an orthogonal projection, hence $0 \leq \mathbf{P}_{V_{D_4}} \leq \text{id}$. The identity on X_n^1 is trivially $0 \leq \text{id} \leq \text{id}$. Combining the blocks: $0 \leq C_n \leq \text{id}$. $\square$

Remark 2.10 (Bounds depend on n). The operator bounds in Proposition 2.9 involve $\deg(G_n^\bullet)$, which grows with n as the refinement subdivides edges and centroid vertices acquire degree at least the lengths of their parent face boundaries. This is expected and does not affect the finite-level theorems of this paper. No inverse-limit boundedness statement for these operators is proved here; in particular, P3 constructs only a vertex inverse-limit Hilbert space ([2, Theorem thm:Xinv-Hilbert]) and explicitly does not construct an edge inverse-limit Hilbert space ([2, Remark rmk:edge-inv]).

3. QUADRATIC CLOSURE FUNCTIONAL AND GRADIENT OPERATOR

3.1. The three quadratic pieces.

Definition 3.1 (Curvature, kinetic, mass quadratic forms). For $\Phi \in X_n$, define:

$$\begin{aligned} R_n(\Phi) &:= \frac{1}{2} \langle \Phi, A_n \Phi \rangle_{X_n} \quad (\text{curvature / graph-Laplacian form}), \\ E_n(\Phi) &:= \frac{1}{2} \langle \Phi, B_n \Phi \rangle_{X_n} \quad (\text{kinetic / coboundary form}), \\ Q_n(\Phi) &:= \frac{1}{2} \langle \Phi, C_n \Phi \rangle_{X_n} \quad (\text{mass form}). \end{aligned}$$

R_n and Q_n are positive semi-definite on X_n (by positive semi-definiteness of A_n and C_n). E_n is indefinite in general (since B_n is self-adjoint but not positive semi-definite: it has pairs of eigenvalues $\pm\sqrt{\lambda}$ for each $\lambda \geq 0$ in $\text{spec}(d_n^* d_n)$).

3.2. The closure functional.

Definition 3.2 (Closure functional F_n). Fix real constants $\alpha, \beta \geq 0, \gamma > 0$. The default working regime, used unless explicitly overridden, is $\alpha, \beta, \gamma > 0$. The *mass-only specialization* is the case $\alpha = \beta = 0, \gamma > 0$. The *closure functional at level n* is

$$F_n(\Phi) := \alpha R_n(\Phi) + \beta E_n(\Phi) - \gamma Q_n(\Phi),$$

a continuous quadratic form on X_n .

Theorem 3.3 (Gradient operator). *For every $n \geq 0$:*

- (1) F_n is a continuous symmetric quadratic form on X_n , equivalently $F_n(\Phi) = \frac{1}{2} \langle \Phi, L_n \Phi \rangle_{X_n}$ for a unique bounded self-adjoint operator $L_n \in B(X_n)$.
- (2) The gradient of F_n at $\Phi \in X_n$ is $\nabla F_n(\Phi) = L_n \Phi$.
- (3) $L_n = \alpha A_n + \beta B_n - \gamma C_n$, with $\|L_n\|_{\text{op}} \leq \alpha \cdot 4 \deg(G_n^\bullet) + \beta \cdot 2\sqrt{\deg(G_n^\bullet)} + \gamma$.

Proof. (1) *Self-adjoint representation.* By Proposition 2.9, A_n, B_n, C_n are each bounded self-adjoint on X_n . Their real linear combination $L_n := \alpha A_n + \beta B_n - \gamma C_n$ is therefore bounded self-adjoint. By the finite-dimensional version of the Riesz representation theorem for quadratic forms [4, Prop. I.3.14], every continuous symmetric quadratic form on a finite-dimensional Hilbert space is uniquely represented as $\frac{1}{2} \langle \Phi, L \Phi \rangle$ for a bounded self-adjoint L . The quadratic form F_n is manifestly continuous (polynomial in matrix entries) and symmetric (each of R_n, E_n, Q_n is symmetric), so this representation exists. By uniqueness, $L_n = \alpha A_n + \beta B_n - \gamma C_n$ is the representing operator.

(2) *Gradient.* For any bounded self-adjoint L on a finite-dimensional Hilbert space, the quadratic form $Q(\Phi) := \frac{1}{2} \langle \Phi, L \Phi \rangle$ is Fréchet differentiable with derivative $\nabla Q(\Phi) = L \Phi$ [5, §4.6].

(3) *Operator-norm bound.* By the triangle inequality:

$$\|L_n\|_{\text{op}} \leq \alpha \|A_n\|_{\text{op}} + \beta \|B_n\|_{\text{op}} + \gamma \|C_n\|_{\text{op}} \leq \alpha \cdot 4 \deg(G_n^\bullet) + \beta \cdot 2\sqrt{\deg(G_n^\bullet)} + \gamma,$$

using Proposition 2.9. □

Remark 3.4 (Sign of L_n). $L_n = \alpha A_n + \beta B_n - \gamma C_n$ is *not* in general positive semi-definite: the indefinite term βB_n and the subtracted mass term $-\gamma C_n$ can produce negative eigenvalues. Whether $L_n \geq 0$ depends on the specific (α, β, γ) and on spectral properties of the constituent operators. We will *not* assume $L_n \geq 0$ in general; the continuous-time flow below is defined and studied for arbitrary bounded self-adjoint L_n , and the norm-contractivity statements are restricted to coercive subspaces where positivity holds locally.

4. CONTINUOUS-TIME GRADIENT FLOW

Theorem 4.1 (Gradient flow exists and is unique). *For every initial datum $\Phi_{n,0} \in X_n$, the ordinary differential equation*

$$(1) \quad \dot{\Phi}_n(t) = -L_n \Phi_n(t), \quad \Phi_n(0) = \Phi_{n,0}$$

has a unique global-in-time C^1 solution $\Phi_n: [0, \infty) \rightarrow X_n$ given by

$$\Phi_n(t) = e^{-tL_n} \Phi_{n,0},$$

where $e^{-tL_n} := \sum_{k=0}^{\infty} \frac{(-t)^k L_n^k}{k!}$ is the finite-dimensional matrix exponential of $-tL_n$. The map $e^{-tL_n}: X_n \rightarrow X_n$ is a bounded linear operator with $\|e^{-tL_n}\|_{\text{op}} \leq e^{t\|L_n\|_{\text{op}}}$ for every $t \geq 0$ (and $\|e^{-tL_n}\|_{\text{op}} \leq e^{|t|\|L_n\|_{\text{op}}}$ for every $t \in \mathbb{R}$ when the extension to negative times is used; Cor. 4.2), and the family $(e^{-tL_n})_{t \geq 0}$ is a C^0 -semigroup on X_n : $e^{0 \cdot L_n} = \text{id}$ and $e^{-(t+s)L_n} = e^{-tL_n} e^{-sL_n}$ for $t, s \geq 0$.

Proof. L_n is a bounded linear operator on a finite-dimensional Hilbert space, so the linear ODE (1) has a unique solution for every initial datum, by the standard Picard–Lindelöf theorem for linear ODEs with bounded coefficients [7, Theorem 1.1]; the solution is given by the matrix exponential $\Phi_n(t) = e^{-tL_n} \Phi_{n,0}$ [5, §6.2]. The matrix exponential has operator norm bounded by $e^{t\|L_n\|_{\text{op}}}$ (direct series estimate); on a finite-dimensional space the series converges in operator norm. Semigroup property: by the functional-equation identity $e^A e^B = e^{A+B}$ for commuting A, B , taking $A = -tL_n$ and $B = -sL_n$ gives $e^{-(t+s)L_n}$. $\square$

Corollary 4.2 (Flow is reversible). *The flow e^{-tL_n} extends to all $t \in \mathbb{R}$, forming a one-parameter group of bounded operators. In particular, the flow is invertible on X_n .*

Proof. $(e^{-tL_n})^{-1} = e^{tL_n}$ by direct computation. The time-reversed ODE $\dot{\Phi}_n(s) = L_n \Phi_n(s)$ has the same well-posedness theorem applied to $-L_n$ in place of L_n . $\square$

Remark 4.3 (Finite-dimensional semigroup). Because X_n is finite-dimensional, the distinction between “bounded semigroup” and “ C_0 -semigroup” that matters in infinite-dimensional PDE theory is absent here. The flow e^{-tL_n} is simply a matrix exponential, and all analytic properties (continuity in t , differentiability, strong/uniform topology equivalence) follow from finite-dimensional linear algebra.

5. ENERGY DISSIPATION

Proposition 5.1 (Energy dissipation identity). *For every initial datum $\Phi_{n,0} \in X_n$ and every $t \geq 0$, with $\Phi_n(t) = e^{-tL_n} \Phi_{n,0}$:*

$$\frac{d}{dt} F_n(\Phi_n(t)) = -\|L_n \Phi_n(t)\|_{X_n}^2 \leq 0.$$

Hence $F_n(\Phi_n(t))$ is non-increasing in t , and the integrated dissipation identity holds:

$$F_n(\Phi_{n,0}) - F_n(\Phi_n(t)) = \int_0^t \|L_n \Phi_n(s)\|_{X_n}^2 ds.$$

Proof. Using the gradient formula $\nabla F_n(\Phi) = L_n \Phi$ (Theorem 3.3(2)) and the flow equation $\dot{\Phi}_n = -L_n \Phi_n$:

$$\begin{aligned} \frac{d}{dt} F_n(\Phi_n(t)) &= \langle \nabla F_n(\Phi_n(t)), \dot{\Phi}_n(t) \rangle_{X_n} = \langle L_n \Phi_n(t), -L_n \Phi_n(t) \rangle_{X_n} \\ &= -\|L_n \Phi_n(t)\|_{X_n}^2. \end{aligned}$$

The sign is ≤ 0 since $\|\cdot\|^2 \geq 0$. Integrating over $[0, t]$ gives the integrated identity. $\square$

Remark 5.2 (Energy decay vs norm decay: distinct theorems). Proposition 5.1 says F_n decreases along the gradient flow. It does *not* say $\|\Phi_n(t)\|_{X_n}$ decreases. In general these are different statements:

- Energy monotonicity holds unconditionally (since $-\|L_n \Phi_n(t)\|^2 \leq 0$).
- Norm contractivity $\|\Phi_n(t)\|_{X_n} \leq \|\Phi_n(0)\|_{X_n}$ holds iff e^{-tL_n} is a contraction for all $t \geq 0$, which requires $L_n \geq 0$ (Corollary 5.3 below).

These are separate hypotheses; conflating them would invalidate the paper.

Corollary 5.3 (Norm contractivity under positivity). *If $L_n \geq 0$ (i.e., L_n is positive semi-definite), then $\|e^{-tL_n}\|_{\text{op}} \leq 1$ for all $t \geq 0$, and hence $\|\Phi_n(t)\|_{X_n} \leq \|\Phi_{n,0}\|_{X_n}$ for all $t \geq 0$.*

Proof. For self-adjoint L_n with $L_n \geq 0$, the spectral theorem gives $L_n = \int \lambda dE(\lambda)$ with $\lambda \geq 0$ on the spectral support, so $e^{-tL_n} = \int e^{-t\lambda} dE(\lambda)$ with $|e^{-t\lambda}| \leq 1$ for $\lambda \geq 0$ and $t \geq 0$. Hence $\|e^{-tL_n}\|_{\text{op}} \leq 1$ [4, Thm. VIII.5].

If L_n has any negative eigenvalue $-\mu < 0$, the corresponding eigenvector grows as $e^{t\mu}$, violating contractivity. $\square$

6. DISCRETE-TIME EULER STEP

6.1. The Euler iteration.

Definition 6.1 (Euler operator). For $n \geq 0$ and step size $\eta > 0$, the *Euler operator* is

$$G_{n,\eta} := I - \eta L_n \in B(X_n).$$

The *Euler iteration* is the discrete-time dynamics

$$\Phi_{n,k+1} := G_{n,\eta} \Phi_{n,k} = (I - \eta L_n) \Phi_{n,k}, \quad k = 0, 1, 2, \dots,$$

with initial datum $\Phi_{n,0} \in X_n$.

6.2. Energy monotonicity under small step size.

Proposition 6.2 (Euler step is energy-decreasing for small η). *Let $\Phi_{n,k} \in X_n$ with $L_n \Phi_{n,k} \neq 0$, and set $a := \|L_n \Phi_{n,k}\|^2 > 0$ and $b := \langle L_n \Phi_{n,k}, L_n^2 \Phi_{n,k} \rangle_{X_n} \in \mathbb{R}$. The exact per-step energy change is*

$$F_n(\Phi_{n,k+1}) - F_n(\Phi_{n,k}) = -\eta a + \frac{1}{2} \eta^2 b.$$

Hence:

- (1) If $b \leq 0$, every $\eta > 0$ strictly decreases F_n .
- (2) If $b > 0$, every η with $0 < \eta < 2a/b$ strictly decreases F_n , with the optimum at $\eta^* = a/b$ giving $F_n(\Phi_{n,k+1}) - F_n(\Phi_{n,k}) = -a^2/(2b) < 0$.

In particular, there exists $\eta_0(\Phi_{n,k}) > 0$ such that $F_n(\Phi_{n,k+1}) < F_n(\Phi_{n,k})$ for every $0 < \eta < \eta_0(\Phi_{n,k})$.

Proof. Direct computation:

$$\begin{aligned} F_n(\Phi_{n,k+1}) &= \frac{1}{2} \langle \Phi_{n,k} - \eta L_n \Phi_{n,k}, L_n(\Phi_{n,k} - \eta L_n \Phi_{n,k}) \rangle \\ &= \frac{1}{2} \langle \Phi_{n,k}, L_n \Phi_{n,k} \rangle - \eta \langle L_n \Phi_{n,k}, L_n \Phi_{n,k} \rangle + \frac{1}{2} \eta^2 \langle L_n \Phi_{n,k}, L_n^2 \Phi_{n,k} \rangle \\ &= F_n(\Phi_{n,k}) - \eta a + \frac{1}{2} \eta^2 b. \end{aligned}$$

The claimed sign analysis follows from completing the square in the quadratic $-\eta a + \frac{1}{2} \eta^2 b$: when $b \leq 0$ the expression is $\leq -\eta a < 0$ for $\eta > 0$; when $b > 0$ it is negative iff $0 < \eta < 2a/b$, with minimum at $\eta = a/b$. $\square$

Remark 6.3 (b is not always positive). Note that $b := \langle L_n \Phi, L_n^2 \Phi \rangle$ is not an inner-product-type quantity unless $L_n \geq 0$, and can be negative when L_n has negative eigenvalues on the component of $L_n \Phi$. When $b \leq 0$, the quadratic correction only helps the descent; when $b > 0$ the step-size bound $\eta < 2a/b$ is required. The proposition does *not* assume $L_n \geq 0$.

Remark 6.4 (Spectral version on coercive subspace). If $\Phi_{n,k} \in Y_n$ lies in a coercive invariant subspace with $m_n \text{id}_{Y_n} \leq L_n|_{Y_n} \leq M_n \text{id}_{Y_n}$, then, setting $v := L_n \Phi_{n,k} \in Y_n$, we have $a = \|v\|^2$ and $b = \langle v, L_n v \rangle$, so the spectral bound $m_n \|v\|^2 \leq \langle v, L_n v \rangle \leq M_n \|v\|^2$ yields $m_n a \leq b \leq M_n a$. The pointwise step-size bound $0 < \eta < 2a/b$ therefore holds uniformly on Y_n for any η with $0 < \eta < 2/M_n$. The pointwise optimum $\eta = a/b$ depends on $\Phi_{n,k}$; for a *uniform* choice that minimises the worst-case norm-contraction rate one takes $\eta = 2/(m_n + M_n)$ (cf. Theorem 6.6), but this is not in general the pointwise energy-descent optimum.

6.3. Norm contractivity on coercive invariant subspaces.

Definition 6.5 (Coercive invariant subspace). A subspace $Y_n \subseteq X_n$ is *coercive invariant* for L_n if:

- (1) $L_n Y_n \subseteq Y_n$ (invariance), and
- (2) there exist constants $0 < m_n \leq M_n < \infty$ such that $m_n \text{id}_{Y_n} \leq L_n|_{Y_n} \leq M_n \text{id}_{Y_n}$ (coercivity with spectral gap $m_n > 0$).

Theorem 6.6 (Strict contraction on coercive invariant subspaces). *Suppose $Y_n \subseteq X_n$ is a coercive invariant subspace for L_n with constants $0 < m_n \leq M_n$. Then for every step size η with $0 < \eta < 2/M_n$, the Euler operator $G_{n,\eta} = I - \eta L_n$ is a strict contraction on Y_n :*

$$\|G_{n,\eta}|_{Y_n}\|_{\text{op}} \leq \max(|1 - \eta m_n|, |1 - \eta M_n|) < 1.$$

The optimal contraction rate $\|G_{n,\eta}\|_{\text{op}}$ is minimised at $\eta^* = 2/(m_n + M_n)$, giving $\|G_{n,\eta^*}|_{Y_n}\|_{\text{op}} = (M_n - m_n)/(M_n + m_n) =: \rho_n < 1$.

Proof. On Y_n , L_n is a self-adjoint operator with spectrum in the interval $[m_n, M_n]$. The operator $I - \eta L_n$ on Y_n has spectrum in $[1 - \eta M_n, 1 - \eta m_n]$. For $0 < \eta < 2/M_n$, both endpoints have $|1 - \eta \lambda| < 1$ iff $0 < \eta < 2/\lambda$. The operator norm on Y_n equals the maximum of $|1 - \eta \lambda|$ over $\lambda \in [m_n, M_n]$, which is the larger of $|1 - \eta m_n|$ and $|1 - \eta M_n|$. The optimum is achieved when these two are equal in magnitude and opposite in sign: $1 - \eta m_n = -(1 - \eta M_n)$, giving $\eta = 2/(m_n + M_n)$ and the stated rate. $\square$

Proposition 6.7 (No global contraction without coercivity). *If L_n has any negative eigenvalue on X_n , then $G_{n,\eta} = I - \eta L_n$ is not a contraction on X_n for any $\eta > 0$: there exists $\xi \in X_n$ with $\|G_{n,\eta}\xi\| > \|\xi\|$.*

Proof. If $L_n \xi_0 = -\mu \xi_0$ for some unit eigenvector $\xi_0 \in X_n$ and $\mu > 0$, then $G_{n,\eta} \xi_0 = (1 + \eta \mu) \xi_0$, which has norm $1 + \eta \mu > 1$. $\square$

Remark 6.8 (The error we are not making). Propositions 6.2 and 6.7 together say: the Euler iteration can be energy-decreasing (for small η , on vectors where L_n is active) and norm-growing (on vectors in the negative eigenspace of L_n). These are compatible: energy F_n and norm $\|\Phi\|$ are genuinely different functionals, and L_n can decrease F_n on one eigenvector while increasing $\|\Phi\|$ on another. An earlier cascade-correspondence manuscript claimed $G_{n,\eta}$ is a contraction on X_n for $0 < \eta \leq \|L_n\|^{-1}$ without any positivity hypothesis; that claim is false, as shown by Proposition 6.7.

7. REFINEMENT COMPATIBILITY AND SYMMETRY INTERTWINING

7.1. Refinement compatibility hypothesis.

Definition 7.1 (Refinement-compatible operator family). A family $(T_n)_{n \geq 0}$ of bounded operators $T_n \in B(X_n)$ is *refinement-compatible* if, for every $n \geq 0$:

$$p_{n+1,n} T_{n+1} = T_n p_{n+1,n} \quad \text{on } X_{n+1},$$

where $p_{n+1,n}: X_{n+1} \rightarrow X_n$ is the downward bonding map of [2, Theorem `thm:bonding`] (extended to the direct-sum space $X_n = X_n^0 \oplus X_n^1$ in the natural componentwise way: $p_{n+1,n}(f, A) := (p_{n+1,n}^0 f, p_{n+1,n}^1 A)$).

7.2. Refinement compatibility of the constituent operators. We now need explicit verification that the constituent operators A_n, B_n, C_n form refinement-compatible families. This is a level-by-level claim about the specific coboundary, Laplacian, and projection operators of §2.

Proposition 7.2 (Coboundary refinement-compatibility, with factor $\frac{1}{2}$). *The coboundary family $(d_n)_n$ satisfies*

$$p_{n+1,n}^1 d_{n+1} = \frac{1}{2} d_n p_{n+1,n}^0 \quad \text{on } X_{n+1}^0,$$

where $p_{n+1,n}^1$ is the parent-fibre averaging bonding map of [2, Definition `def:p1`] (using the edge-parent map of Definition `def:edge-parent`), $p_{n+1,n}^0$ is the restriction of [2, Definition `def:p0`],

and the level- $(n+1)$ edges subdivide the level- n edges via the canonical edge-parent map of [2, Definition `def:p1` (using the edge-parent map of Definition `def:edge-parent`)].

Proof. For $f \in X_{n+1}^0$ and $e \in E^{\text{or}}(G_n^\bullet)$:

$$(d_n(p_{n+1,n}^0 f))(e) = (p_{n+1,n}^0 f)(\text{head}(e)) - (p_{n+1,n}^0 f)(\text{tail}(e)) = f(\text{head}(e)) - f(\text{tail}(e))$$

(using $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ and $p^0 = \text{restriction from [2, Definition `def:p0`]}$).

On the other side, the level- $(n+1)$ edge e subdivides as e_1, e_2 with $\text{head}(e_1) = \text{tail}(e_2) = m$ (the midpoint of e), so

$$\begin{aligned} (d_{n+1}f)(e_1) &= f(m) - f(\text{tail}(e)), \\ (d_{n+1}f)(e_2) &= f(\text{head}(e)) - f(m). \end{aligned}$$

The parent-fibre average of $d_{n+1}f$ from [2, Definition `def:p1` (using the edge-parent map of Definition `def:edge-parent`)] is

$$(p_{n+1,n}^1 d_{n+1}f)(e) = \frac{1}{2}((d_{n+1}f)(e_1) + (d_{n+1}f)(e_2)) = \frac{1}{2}(f(\text{head}(e)) - f(\text{tail}(e))) = \frac{1}{2}(d_n p_{n+1,n}^0 f)(e),$$

which proves the claimed factor- $\frac{1}{2}$ identity. $\square$

Remark 7.3 (The factor- $\frac{1}{2}$ is structural). The factor of $\frac{1}{2}$ in Proposition 7.2 is intrinsic to the combination of an *undivided-difference coboundary* d_n at every refinement level with the *averaging bonding map* $p_{n+1,n}^1$ of [2, Definition `def:p1` (using the edge-parent map of Definition `def:edge-parent`)]. Changing the coboundary normalization (e.g. scaling d_n by 2^n or by an edge-length factor) can convert the factor into different constants, but cannot eliminate it without sacrificing some other structural property (e.g. $\|p_{n+1,n}^1\|_{\text{op}} \leq 1$). We keep the factor of $\frac{1}{2}$ explicit and propagate it through the downstream operator identities.

Proposition 7.4 (Mass-block refinement compatibility). *Under the conventions of §2 and [2], the mass block satisfies*

$$p_{n+1,n} C_{n+1} = C_n p_{n+1,n}$$

exactly (no scaling factor). The analogous exact identity for the coboundary, adjoint coboundary, or graph-Laplacian blocks is not established in this paper; see Remark 7.5 below for the scope of what is and is not proved.

Proof. $C_n = \begin{pmatrix} \mathbf{P}_{V_{D_4}} & 0 \\ 0 & \text{id} \end{pmatrix}$ acts pointwise on each vertex / edge value by a fixed linear operation on fibres: on X_n^0 by the projection $\mathbf{P}_{V_{D_4}}: F^0 \rightarrow V_{D_4}$, and on X_n^1 by the identity. Both of these commute with fibrewise restriction (vertex case, $p_{n+1,n}^0$) and with fibrewise parent-fibre averaging (edge case, $p_{n+1,n}^1$). Hence $p_{n+1,n} C_{n+1} = C_n p_{n+1,n}$ exactly. $\square$

Remark 7.5 (Scope of refinement compatibility). What is actually proved here:

- The mass block C_n is refinement-compatible exactly: $p C_{n+1} = C_n p$ (Proposition 7.4).
- The coboundary obeys the factor- $\frac{1}{2}$ relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ (Proposition 7.2).

What is *not* proved here: an analogous factor- $\frac{1}{2}$ relation for d_n^* ; a block identity of the form $p B_{n+1} = c B_n p$ for any explicit constant c ; an induced factor on the graph-Laplacian block A_n . Each of these would require a separate calculation reconciling the distinct p^0 and p^1 bonding-map factors across the off-diagonal blocks, and none is carried out in this paper.

Consequently, we restrict Theorem 7.6 (and Theorem 7.7 below) to the mass-only regime $\alpha = \beta = 0$, where refinement compatibility is positively established. For arbitrary (α, β, γ) we make *no* claim about refinement-intertwining of L_n : neither that it holds (no such calculation is offered), nor that it fails (no negative example is offered). Downstream papers requiring refinement-intertwining outside the mass-only regime must supply their own analysis of the coboundary-adjoint block identities.

7.3. The refinement-intertwining theorem: mass-only case.

Theorem 7.6 (L_n is refinement-compatible in the mass-only case). *Under the restriction $\alpha = \beta = 0$ (i.e., $L_n = -\gamma C_n$), the gradient operator family $(L_n)_{n \geq 0}$ is refinement-compatible:*

$$p_{n+1,n} L_{n+1} = L_n p_{n+1,n} \quad \text{for every } n \geq 0.$$

For arbitrary (α, β, γ) we make no claim about the corresponding identity; see Remark 7.5.

Proof. Under $\alpha = \beta = 0$, $L_n = -\gamma C_n$. By Proposition 7.4, $p_{n+1,n} C_{n+1} = C_n p_{n+1,n}$, hence (multiplying by $-\gamma$) $p_{n+1,n} L_{n+1} = L_n p_{n+1,n}$. $\square$

Theorem 7.7 (Flow intertwining: mass-only case). *Under the restriction $\alpha = \beta = 0$, the continuous-time flow of L_n intertwines with the bonding maps:*

$$p_{n+1,n} e^{-tL_{n+1}} = e^{-tL_n} p_{n+1,n} \quad \text{for every } t \geq 0.$$

For arbitrary (α, β, γ) we make no claim about the corresponding identity; see Remark 7.5.

Proof. Under $\alpha = \beta = 0$, Theorem 7.6 gives $p_{n+1,n} L_{n+1} = L_n p_{n+1,n}$ exactly. By functional calculus for bounded operators on finite-dimensional Hilbert spaces [4, Thm. VIII.5], applied to the power series $e^{-tx} = \sum_{k \geq 0} (-tx)^k / k!$ term by term (each power L_n^k commutes with $p_{n+1,n}$ by iteration of the linear intertwining), we obtain $p_{n+1,n} e^{-tL_{n+1}} = e^{-tL_n} p_{n+1,n}$. $\square$

Remark 7.8 (Pedagogical note on §7 scope). Section 7 *does not* establish refinement-intertwining of the full cascade gradient flow with arbitrary α, β, γ . It establishes:

- an explicit factor- $\frac{1}{2}$ relation for the coboundary (Proposition 7.2);
- exact compatibility for the mass block (Proposition 7.4);
- exact compatibility for L_n and its flow in the mass-only degenerate case (Theorems 7.6, 7.7).

Any downstream paper needing a refinement-intertwining theorem for non-degenerate (α, β, γ) must supply its own analysis. This paper makes no claim about whether such intertwining holds in that regime; it is simply not established here.

7.4. Symmetry intertwining.

Corollary 7.9 (Symmetries commuting with L_n commute with the flow). *If $g \in W^\bullet$ acts on X_n by $\rho_n(g) \in B(X_n)$ with $\rho_n(g) L_n = L_n \rho_n(g)$ (i.e., g -equivariant gradient operator), then $\rho_n(g) e^{-tL_n} = e^{-tL_n} \rho_n(g)$ for every $t \geq 0$.*

Proof. Same functional-calculus argument as Theorem 7.7. $\square$

8. SECTORWISE INTERTWINERS: EXPLICIT BOUNDS

8.1. Admissible finite-level intertwiners.

Definition 8.1 (Generator list). The *finite-level generator list* $\mathcal{G}_n$ at level n consists of the following bounded real-Hilbert-space operators:

- downward bonding maps: $p_{n+1,n}^0, p_{n+1,n}^1$ and their composition $p_{n+1,n}$ on $X_{n+1} \rightarrow X_n$;
- upward cylindrical embeddings: $j_{n,n+1}^0, j_{n,n+1}^1$ and $j_{n,n+1}$ on $X_n \rightarrow X_{n+1}$;
- sector projections: $\mathbf{P}_{V_{H_4}}, \mathbf{P}_{V_{D_4}}, \mathbf{P}_{V_{16}}, \mathbf{P}_{V_8}$ on X_n^0 ;
- Coxeter actions: $\rho_n(g)$ for $g \in W^\bullet$;
- discrete differential operators: $d_n, d_n^*, A_n, B_n, C_n, L_n$ from §2;
- flow-operator family: $\{e^{-tL_n} : t \geq 0\}$;
- the G_2 -stabilizer fibre map $\text{Im}(\mathbb{O}) \rightarrow \text{Im}(\mathbb{O})$ fixing a chosen unit i_0 (an orthogonal map of norm 1, imported from [1]).

The rationality-side σ -involution and the icosian star-map are *not* elements of $\mathcal{G}_n$: by P3 ([2, Definition def:sigma, Theorem thm:commute]) these are defined only on the mixed-form $\mathbb{Q}(\varphi)/\mathbb{Q}$ vertex space and do not extend to bounded operators on the real Hilbert space X_n^0 . They appear in this paper only at the rational-form level, never as elements of $\mathcal{G}_n$.

Definition 8.2 (Admissible intertwiner, syntactic). An *admissible finite-level intertwiner* is a formal expression obtained from the generator list $\mathcal{G}_n$ (or $\mathcal{G}_n \cup \mathcal{G}_m$ for inter-level intertwiners) by the syntactic rules:

- Every generator $T \in \mathcal{G}_n$ is an admissible intertwiner (of level n , or between appropriate levels).
- If T_1, T_2 are admissible intertwiners with compatible domains and codomains (matching refinement levels and state-space types), then $c_1 T_1 + c_2 T_2$ is an admissible intertwiner for any $c_1, c_2 \in \mathbb{R}$.
- If $T_1: X_m \rightarrow X_\ell$ and $T_2: X_n \rightarrow X_m$ are admissible, then the composition $T_1 \circ T_2: X_n \rightarrow X_\ell$ is admissible.

The definition is *syntactic*: it does not assume that the resulting operator is bounded *a priori*. Boundedness is proved in Theorem 8.4 below from the printed generator bounds.

8.2. Explicit bounds for the named generators.

Proposition 8.3 (Generator bounds table). *The named generators from Definition 8.1 satisfy the following explicit operator-norm bounds on their respective finite-level Hilbert spaces:*

Generator	Norm bound	Source
$p_{n+1,n}^0$ (vertex bonding)	$\ p^0\ _{\text{op}} \leq 1$	[2, Theorem <code>thm:bonding</code>]
$p_{n+1,n}^1$ (edge bonding)	$\ p^1\ _{\text{op}} \leq 1$	[2, Theorem <code>thm:bonding</code>]
$p_{n+1,n}$ (direct-sum bonding $X_{n+1} \rightarrow X_n$)	$\ p\ _{\text{op}} \leq 1$	direct-sum norm, from $\ p^0\ , \ p^1\ $
$j_{n,n+1}^0$ (vertex embedding)	$\ j^0\ _{\text{op}} = 1$	[2, Proposition <code>prop:adjoints</code>]
$j_{n,n+1}^1$ (edge embedding)	$\ j^1\ _{\text{op}} = 2^{-1/2}$	[2, Proposition <code>prop:adjoints</code>]
$j_{n,n+1}$ (direct-sum embedding $X_n \rightarrow X_{n+1}$)	$\ j\ _{\text{op}} = 1$	direct-sum norm: $\max(\ j^0\ , \ j^1\)$
$\mathbf{P}_{V_\bullet}$ (sector projections)	$\ \mathbf{P}\ _{\text{op}} = 1$	orthogonal projections
$\rho_n(g)$ (Coxeter action)	$\ \rho_n(g)\ _{\text{op}} = 1$	unitary, [2, Lemma <code>lem:coxet</code>]
d_n (coboundary)	$\ d_n\ _{\text{op}} \leq 2\sqrt{\deg(G_n^\bullet)}$	Prop. 2.9(1)
d_n^* (adjoint coboundary)	$\ d_n^*\ _{\text{op}} \leq 2\sqrt{\deg(G_n^\bullet)}$	adjoint of d_n
A_n (graph-Laplacian block)	$\ A_n\ _{\text{op}} \leq 4\deg(G_n^\bullet)$	Prop. 2.9(2)
B_n (coboundary block)	$\ B_n\ _{\text{op}} \leq 2\sqrt{\deg(G_n^\bullet)}$	Prop. 2.9(3)
C_n (mass block)	$0 \leq C_n \leq \text{id}$, so $\ C_n\ _{\text{op}} \leq 1$	Prop. 2.9(4)
L_n (gradient operator)	$\ L_n\ _{\text{op}} \leq 4\alpha \deg + 2\beta\sqrt{\deg} + \gamma$	Thm. 3.3(3)
e^{-tL_n} (flow, $t \geq 0$)	$\ e^{-tL_n}\ _{\text{op}} \leq e^{t\ L_n\ _{\text{op}}}$	Thm. 4.1

The G_2 -stabilizer fibre map (from [1, Definitions `def:G2`, `def:stab-u` and Fact `fact:SU3-stab`]) acts on $\text{Im}(\mathbb{O}) = F^1$ as a compact-group element of orthogonal norm 1; its ℓ^2 -extension to X_n^1 therefore has operator norm exactly 1.

Proof. Each row of the table is verified by citation or by the referenced proposition. $\|\rho_n(g)\|_{\text{op}} = 1$ because $\rho_n(g)$ is unitary ([2, Lemma `lem:coxeter-unitary`]: it permutes an ℓ^2 -orthonormal basis up to fibre-level orthogonal transformations). The G_2 -stabilizer claim uses that the relevant element is in the compact orthogonal group of $\text{Im}(\mathbb{O})$. $\square$

Theorem 8.4 (Admissible intertwiners are bounded, continuous, Borel). *Every admissible finite-level intertwiner T (in the syntactic sense of Definition 8.2) is a bounded linear operator on the finite-dimensional Hilbert space in its codomain. Specifically:*

- (1) (Boundedness with printed bound.) If T is expressed as a composition/sum of generators with the bounds of Proposition 8.3, then $\|T\|_{\text{op}}$ is bounded by the corresponding combination of the listed generator bounds (via $\|T_1 \pm T_2\|_{\text{op}} \leq \|T_1\|_{\text{op}} + \|T_2\|_{\text{op}}$ and $\|T_1 T_2\|_{\text{op}} \leq \|T_1\|_{\text{op}} \|T_2\|_{\text{op}}$).
- (2) (Continuity.) T is continuous as a linear map between finite-dimensional Hilbert spaces (automatic for bounded linear maps).
- (3) (Borel measurability.) T is Borel-measurable (continuity plus finite-dimensionality).

- (4) (*Symmetry intertwining, conditional.*) If every generator T_k appearing in the composition $T = T_{k_1} \circ \cdots \circ T_{k_m}$ satisfies a prescribed intertwining relation $\rho(g)T_k = T'_k \rho(g)$ with a symmetry g , then T inherits the same relation: $\rho(g)T = (T'_{k_1} \cdots T'_{k_m})\rho(g)$.

Proof. (1) *Boundedness with printed bound.* By structural induction on the formal expression of T per Definition 8.2, using the sub-multiplicativity of $\|\cdot\|_{\text{op}}$ on compositions and the triangle inequality on sums. Base case: each generator has a norm bound from Proposition 8.3. Inductive cases (sum, composition): standard operator-theoretic inequalities [4, Ch. I.1].

(2) *Continuity.* Bounded linear maps on normed spaces are continuous [4, Thm. I.1.3].

(3) *Borel measurability.* Continuous maps between topological spaces are Borel measurable [9, Cor. 1.2.12]. In finite dimension, all reasonable measurable structures coincide.

(4) *Symmetry intertwining.* By iterated application of the generator intertwining along the composition sequence. $\square$

Remark 8.5 (What this replaces). Theorem 8.4 is the structural replacement for the false global theorem “every correspondence arising from closure dynamics is Borel and equivariant” that appeared in an earlier foundations draft. Instead of one grand theorem, we have:

- a printed generator list (Definition 8.1);
- a printed bounds table (Proposition 8.3);
- a compositionality theorem (Theorem 8.4).

Downstream papers naming an intertwiner as a specific composition of generators read off the operator bound, continuity, and intertwining properties from this table and theorem. No abstract “Borel/equivariance” theorem is invoked; every claim is composable from printed ingredients.

9. CITATION CONTRACT AND DOWNSTREAM HANDOFF

9.1. What downstream papers may cite.

- (1) Definition 3.2 and Theorem 3.3: the closure functional F_n and its gradient operator $L_n = \alpha A_n + \beta B_n - \gamma C_n$, a bounded self-adjoint operator on X_n .
- (2) Theorem 4.1: global existence and uniqueness of the continuous-time gradient flow $\Phi_n(t) = e^{-tL_n}\Phi_n(0)$.
- (3) Proposition 5.1: energy dissipation along the flow.
- (4) Propositions 6.2, 6.7 and Theorem 6.6: discrete-time Euler analysis, with energy monotonicity and norm contractivity cleanly separated.
- (5) Proposition 7.2: the explicit factor- $\frac{1}{2}$ coboundary relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$. Proposition 7.4: exact mass-block compatibility $pC_{n+1} = C_n p$. Theorems 7.6 and 7.7: refinement compatibility of L_n and of the continuous-time flow *in the mass-only case* $\alpha = \beta = 0$. For arbitrary (α, β, γ) this paper makes no claim about refinement-intertwining of L_n ; see Remark 7.5.
- (6) Definitions 8.1, 8.2 and Theorem 8.4: explicit generator-family list and bounded-intertwiner theorem replacing the false global Borel/equivariance theorem.

9.2. Recommended citation phrasing. “By P4 [3], each finite refinement level n carries an explicit quadratic closure functional $F_n = \frac{1}{2}\langle \cdot, L_n \cdot \rangle$ with bounded self-adjoint gradient L_n (Theorem 3.3), continuous-time semigroup flow e^{-tL_n} (Theorem 4.1), energy monotonicity (Proposition 5.1), strict contraction only on coercive invariant subspaces (Theorem 6.6), an explicit factor- $\frac{1}{2}$ coboundary refinement relation (Proposition 7.2) and exact mass-block refinement compatibility (Proposition 7.4), with full refinement-intertwining $p_{n+1,n} e^{-tL_{n+1}} = e^{-tL_n} p_{n+1,n}$ holding in the mass-only case $\alpha = \beta = 0$ (Theorem 7.7); and explicit bounded intertwiner bounds from the explicit generator-family list (Theorem 8.4).”

9.3. What this paper does not provide.

- (1) No continuum-limit statement about $(L_n)_n$ as $n \rightarrow \infty$ or about $(e^{-tL_n})_n$.
- (2) No target-side metric / hydrodynamic / gauge / spectral / Hecke / cohomology / CTM/TM correspondence theorem.

- (3) No PDE identification of $\dot{\Phi}_n = -L_n\Phi_n$ with any classical evolution equation.
- (4) No universal Borel/equivariance principle beyond the explicit explicit generator-family list.

APPENDIX A. NOTATION AUDIT

Symbol	Meaning	Defined
X_n	combined state space $X_n^0 \oplus X_n^1$	Def. 2.1
D_n	domain of the operators ($= X_n$)	Def. 2.1
d_n, d_n^*	discrete coboundary and adjoint	Defs. 2.3, 2.4
A_n	graph-Laplacian block operator	Def. 2.6
B_n	coboundary block operator	Def. 2.7
C_n	mass block operator	Def. 2.8
R_n, E_n, Q_n	curvature, kinetic, mass quadratic forms	Def. 3.1
F_n	closure functional $\alpha R_n + \beta E_n - \gamma Q_n$	Def. 3.2
L_n	gradient operator $\alpha A_n + \beta B_n - \gamma C_n$	Thm. 3.3(3)
e^{-tL_n}	continuous-time flow	Thm. 4.1
$G_{n,\eta}$	Euler operator $I - \eta L_n$	Def. 6.1
Y_n	coercive invariant subspace	Def. 6.5
m_n, M_n	coercive spectral bounds	Def. 6.5
$\mathcal{G}_n$	finite-level generator list	Def. 8.1

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- [1] *Cascade-Classical Correspondence II: The Cascade Algebraic Substrate*, `papers/cascade-algebraic-substrate/cascade-algebraic-substrate.tex`. (Paper P2.) Pinned results used here, cited by label: `def:icosian-star` (icosian star-map $*$ on the icosian ring $I \subset \mathbb{H}$, identified at the rational level with the coefficientwise σ of P1; this map is used in this paper only at the rational-form level, never as a real-Hilbert-space operator on X_n^0); `def:G2`, `def:stab-u`, and `Fact fact:SU3-stab` (G_2 stabilizer of a unit imaginary octonion $\cong \text{SU}(3)$); `def:C1-basis`, `def:octonions`, and `def:Q-forms` (sector-fibre $\mathbb{Q}$ -forms used in P3's mixed-form construction).
- [2] *Cascade-Classical Correspondence III: Refinement Spaces and Symmetry Actions*, `papers/cascade-refinement-spaces/cascade-refinement-spaces.tex`. (Paper P3.) Pinned results used here, cited by label: Notation `not:fibres` (sector fibres F^0, F^1); Definitions `def:Xn-0` and `def:Xn-1` (finite-level ℓ^2 spaces X_n^0, X_n^1); Proposition `prop:Xn-Hilbert` (finite-level Hilbert structure); Definition `def:edge-parent` (canonical edge-parent map on subdividing edges); Definition `def:p0` (vertex bonding p^0 as restriction); Definition `def:p1` (edge bonding p^1 as parent-fibre averaging); Theorem `thm:bonding` (operator-norm contraction bounds for p^0, p^1); Proposition `prop:adjoints` (adjoints $j_{n,n+1} = (p_{n+1,n})^*$, giving $\|j^0\|_{\text{op}} = 1$ and $\|j^1 B\|^2 = \frac{1}{2}\|B\|^2$); Definition `def:coxeter` (branchwise Coxeter action $\rho_n(g)$); Lemma `lem:coxeter-unitary` (unitarity of $\rho_n(g)$ on each $X_n^{0,\bullet}, X_n^{1,\bullet}$); Theorem `thm:Xinv-Hilbert` (vertex inverse-limit Hilbert space; no edge inverse-limit Hilbert space is constructed in P3, see Remark `rmk:edge-inv` of P3).
- [3] (This paper.) *Cascade-Classical Correspondence IV: Discrete Closure Dynamics*.
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- [8] P. Hartman, *Ordinary Differential Equations*, 2nd ed., Birkhäuser, 1982. Linear ODEs: Ch. IV.
- [9] V. I. Bogachev, *Measure Theory*, Vol. I, Springer, 2007. Borel measurability of continuous maps: Cor. 1.2.12.